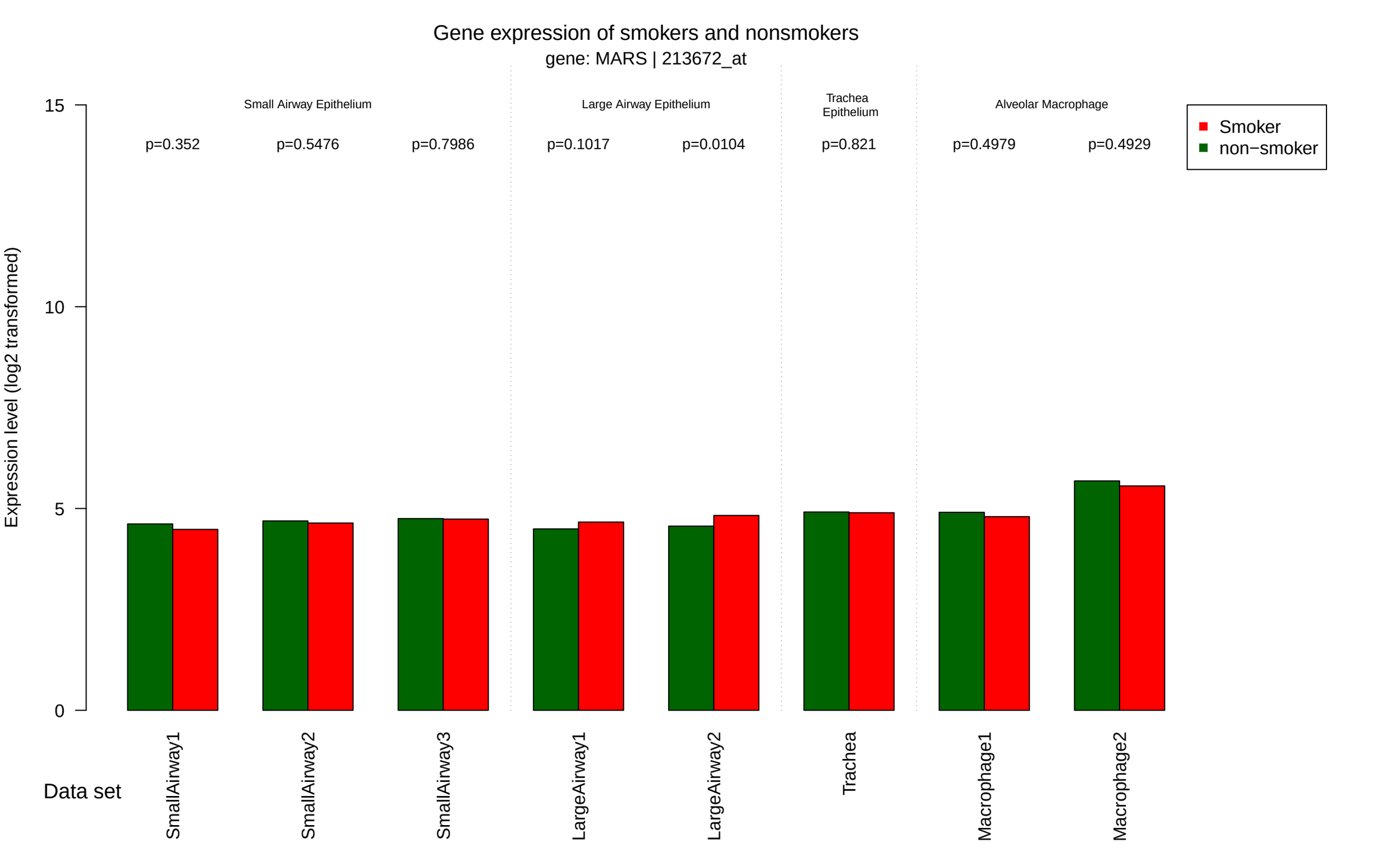
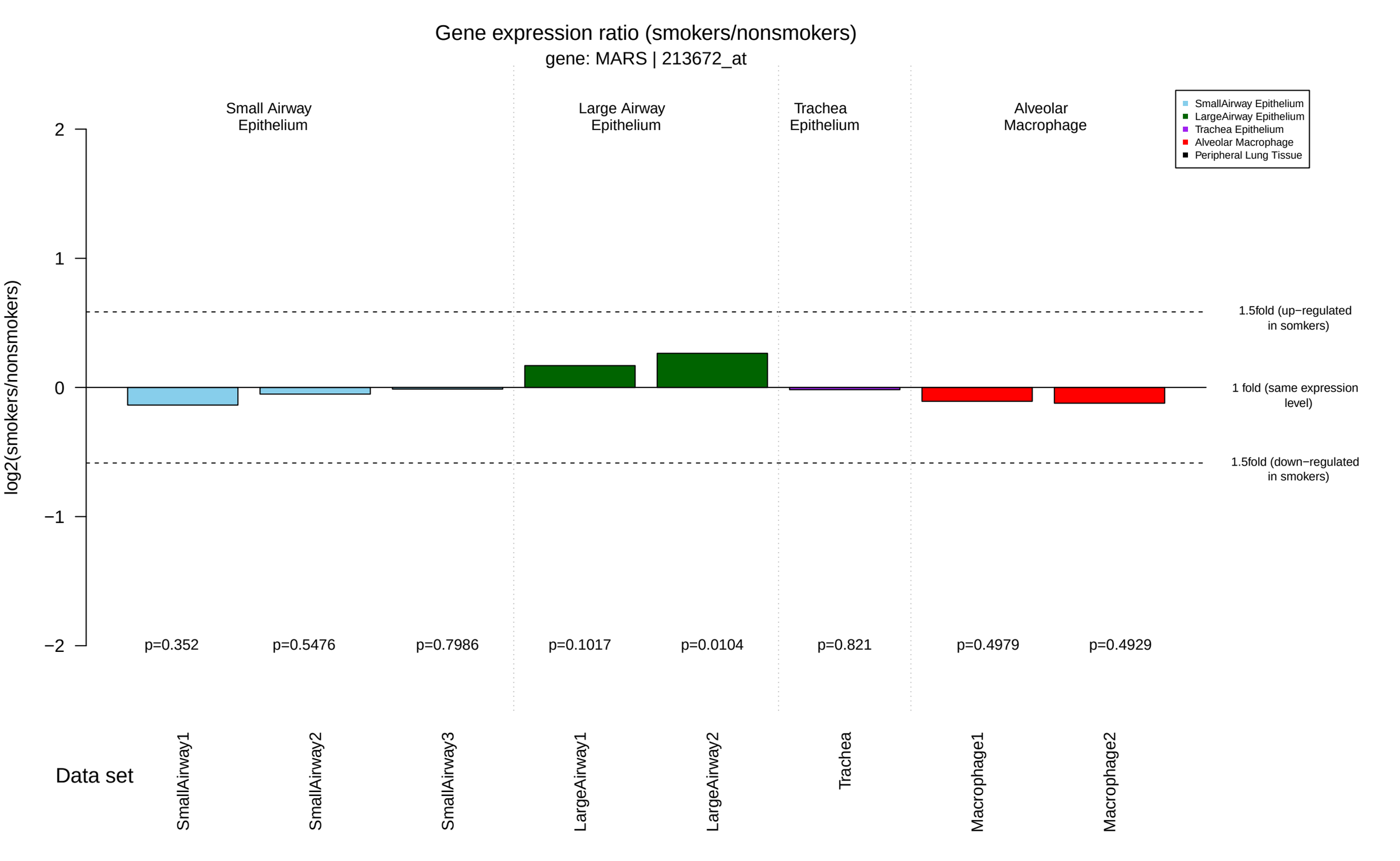
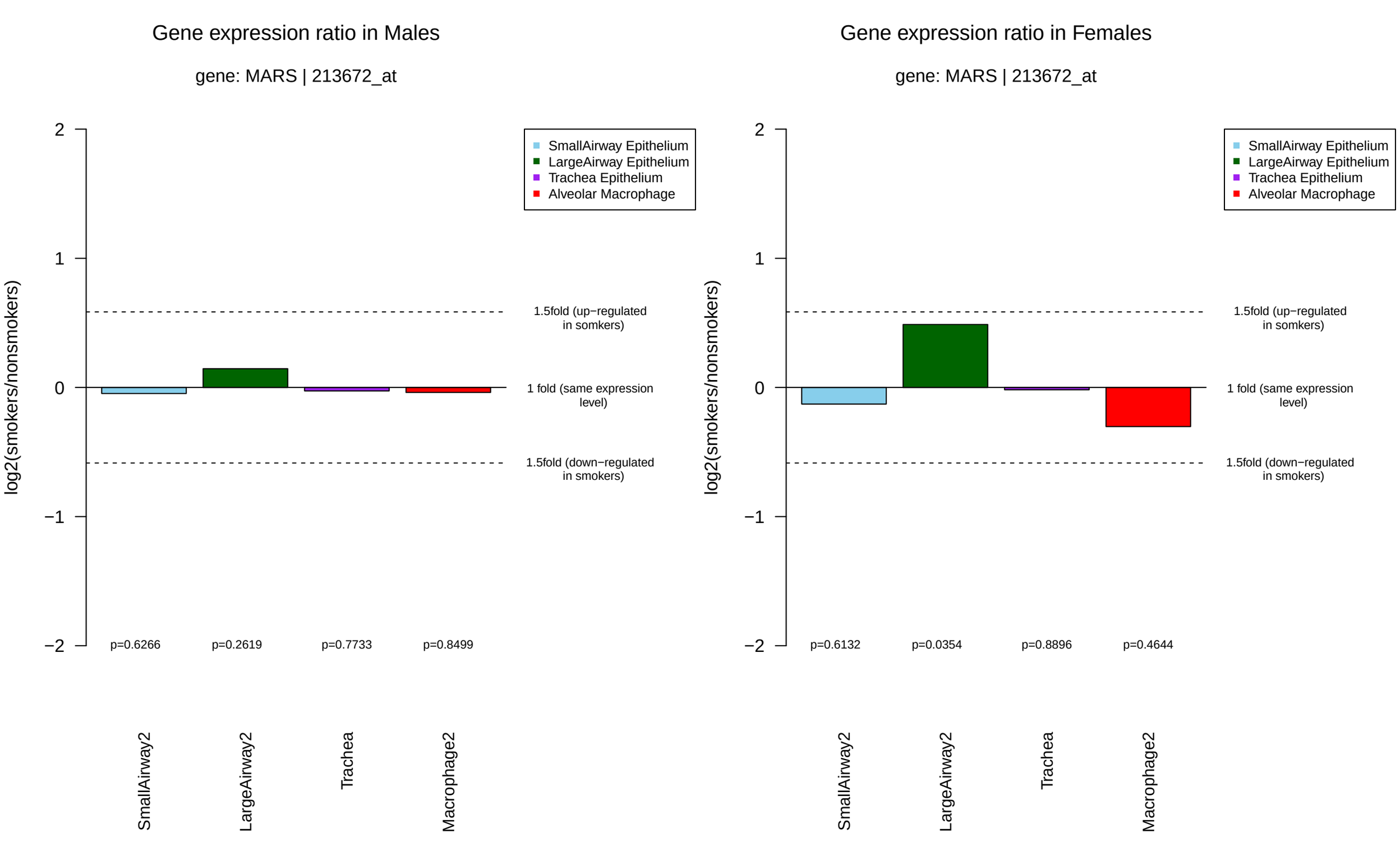
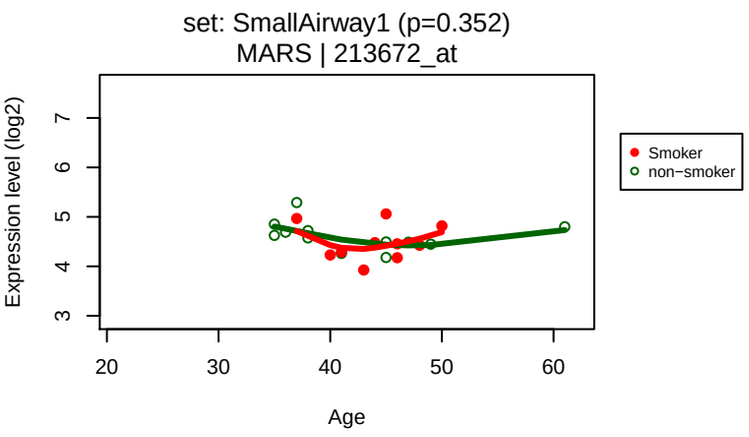
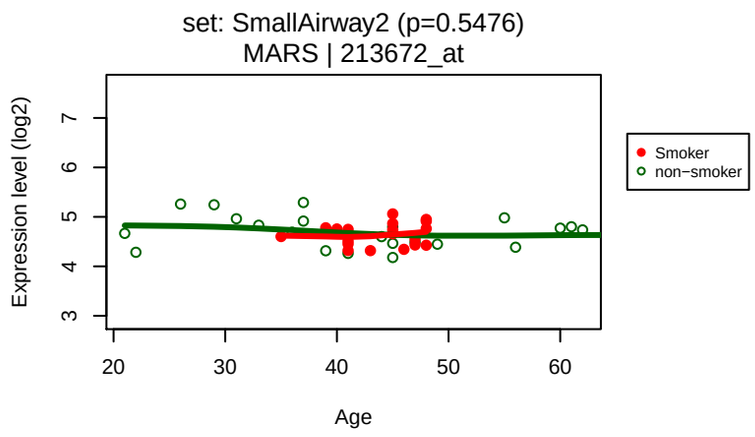
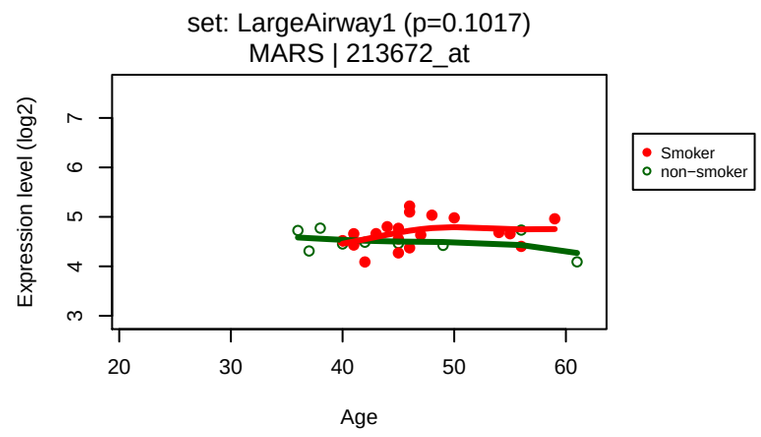

Gene expression of smokers and nonsmokers
gene: MARS | 213672_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.352
p=0.5476
p=0.7986
p=0.1017
p=0.0104
p=0.821
p=0.4979
p=0.4929
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

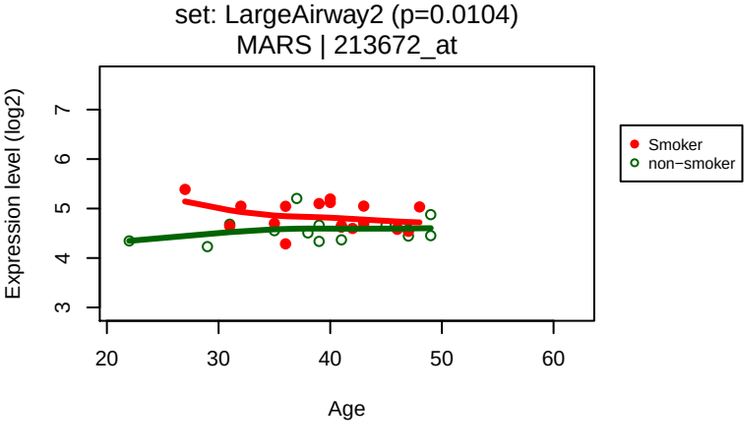

Gene expression ratio (smokers/nonsmokers)
gene: MARS | 213672_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.352
p=0.5476
p=0.7986
p=0.1017
p=0.0104
p=0.821
p=0.4979
p=0.4929
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

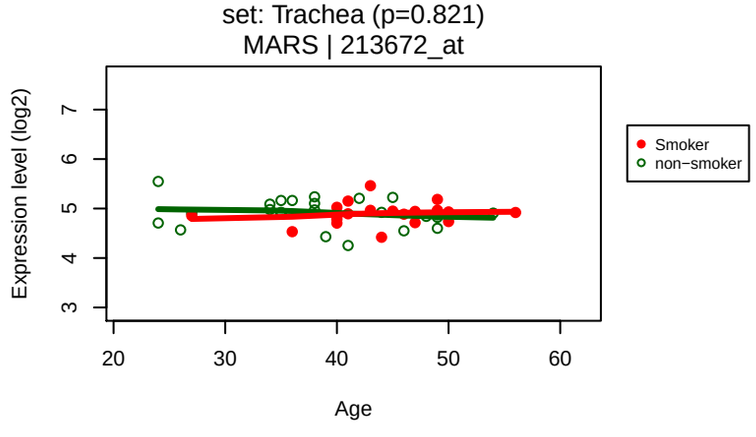

Gene expression ratio in Males
gene: MARS | 213672_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6266
p=0.2619
p=0.7733
p=0.8499
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MARS | 213672_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6132
p=0.0354
p=0.8896
p=0.4644
SmallAirway2
LargeAirway2
Trachea
Macrophage2

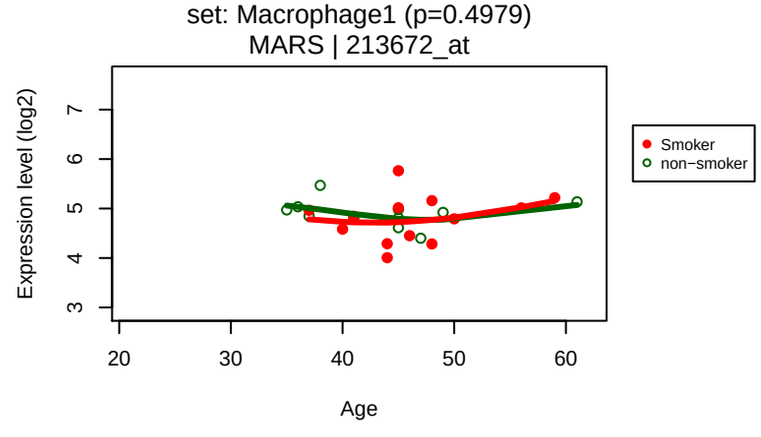

set: SmallAirway1 (p=0.352)
MARS | 213672_at
Expression level (log2)
Age
Smoker
non−smoker

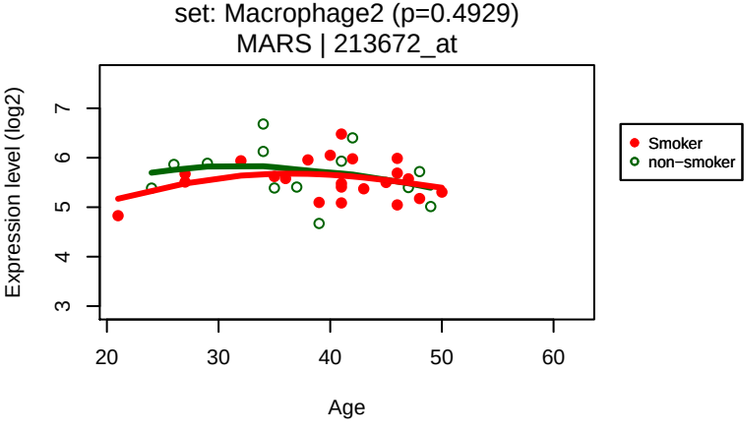

set: SmallAirway2 (p=0.5476)
MARS | 213672_at
Expression level (log2)
Age
Smoker
non−smoker

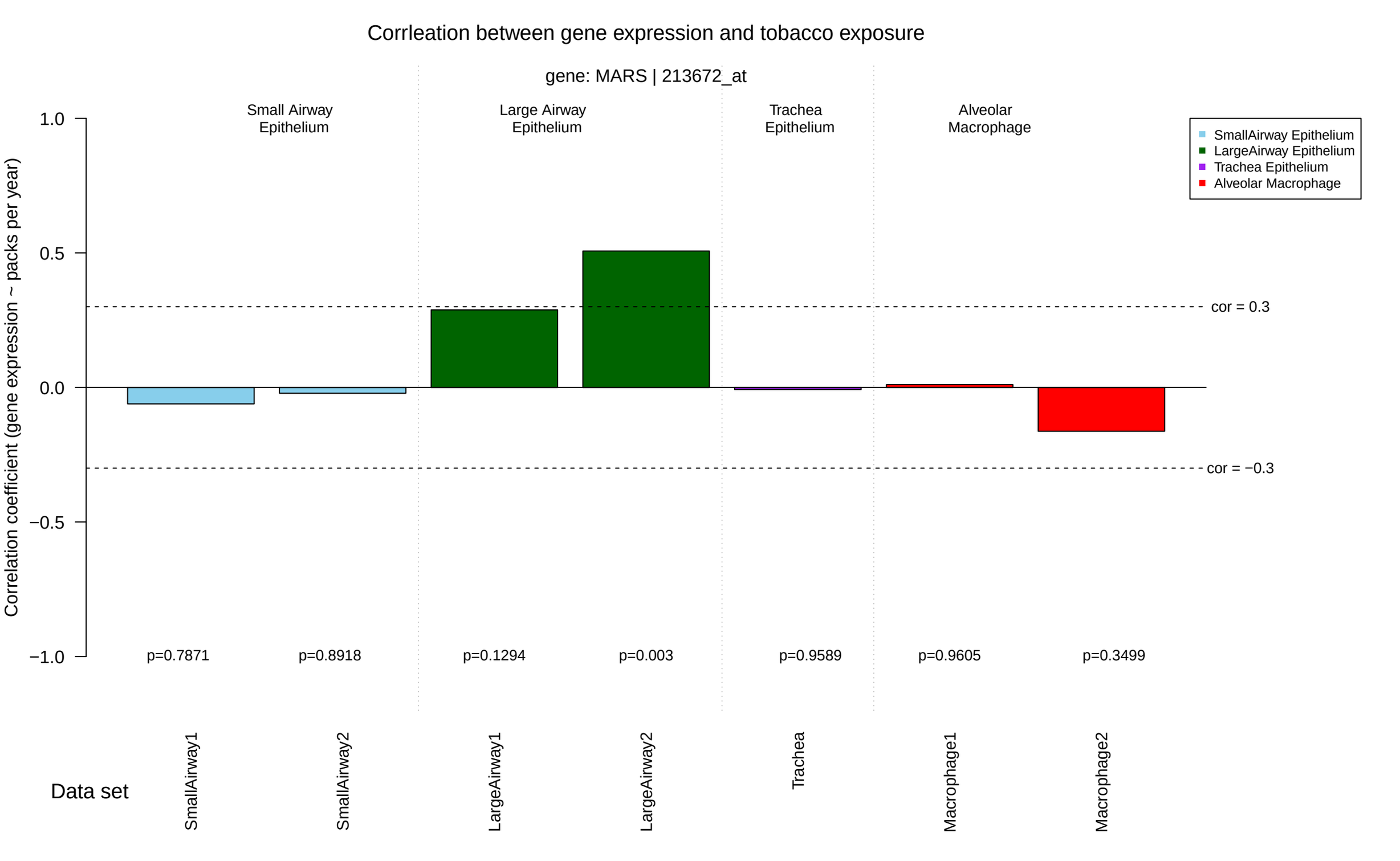

set: LargeAirway1 (p=0.1017)
MARS | 213672_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0104)
MARS | 213672_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.821)
MARS | 213672_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4979)
MARS | 213672_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4929)
MARS | 213672_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MARS | 213672_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7871
p=0.8918
p=0.1294
p=0.003
p=0.9589
p=0.9605
p=0.3499
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2